

4.2)

a) SEE CODE

$$b) \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \times \frac{-b \mp \sqrt{b^2 - 4ac}}{-b \mp \sqrt{b^2 - 4ac}}$$

$$= \frac{b^2 \pm b\sqrt{b^2 - 4ac} \mp b\sqrt{b^2 - 4ac} - b^2 + 4ac}{2a(-b \mp \sqrt{b^2 - 4ac})}$$

$$= \frac{2c}{-b \mp \sqrt{b^2 - 4ac}}$$

SEE CODE

c) SEE CODE.

4.4) a) FIRST COMPUTE THE MEAN

$$\bar{x} = \frac{1}{5} (2x-2 + x-1 + x + x+1 + x+2)$$
$$= \frac{x}{5}$$

THEN, COMPUTE THE MEAN OF THE SQUARE

$$\overline{x^2} = \frac{1}{5} \left[(2x-2)^2 + (x-1)^2 + x^2 + (x+1)^2 + (x+2)^2 \right]$$
$$= \frac{1}{5} \left[x^2 - 4x + 4 + x^2 - 2x + 1 + x^2 + x^2 + 2x + 2 + x^2 + 4x + 4 \right]$$
$$= \frac{1}{5} [5x^2 + 10]$$
$$= x^2 + 2$$

$$\text{VAR } x = \overline{x^2} - \bar{x}^2 = x^2 + 2 - x^2 = 2$$

$$\Rightarrow \boxed{\text{VAR } x = 2}$$

b) SEE CODE

THE RESULT FROM 4.4 IS NOT ACCURATE. THAT'S BECAUSE BOTH $\overline{x^2}$ AND $\bar{x}^2$ ARE LARGE NUMBERS COMPARED TO THE DIFFERENCE BETWEEN THEM.

c) PICK 4.3 TO AVOID PROBLEMS UNLESS YOU KNOW YOU ARE DEALING WITH SMALL NUMBERS.